Google Cloud H2S

Gen AI Academy
APAC Edition

Build in APAC. Build for the world.

PARTICIPANT DETAILS

a. Participant name : **Hariharan C**

b. Problem Statement : **Build an AI agent that uses the Model Context Protocol (MCP) to connect to one external tool or data source, retrieve information, and use that information in its response.**

c. Github repo link : *https://github.com/BlackWizard123/Track-2---Medical_MCP_agent*

d. Cloud Run URL : *https://medical-agent-1059652519537.us-central1.run.app/chat*

Brief about the idea

- **MediChat** is an AI-powered medical assistant agent
- Provides quick, reliable, and structured medical information
- Uses **Google ADK + Gemini 2.5 Flash** for intelligent responses
- Connects to a structured medical knowledge base via **MCP**
- Dynamically retrieves data using specialized tools (disease info & remedies)
- Ensures safe responses with medical disclaimers
- Deployed on **Google Cloud Run** for scalable, real-time access

Meeting the Build Criteria

How did you navigate the requirements of your chosen track to build a custom solution - using ADK, MCP, or AlloyDB AI that effectively addresses the 'What You Must Build' criteria?

- Built the agent using **Google ADK** to manage sessions, tool calling, and response generation
- Created a **custom structured knowledge base (data.json)** containing medical data
- Implemented **two MCP tools**:
    1. ***disease_info*** → symptoms, causes, precautions
    2. ***disease_remedy*** → treatments, when to consult a doctor
- Enabled **intent-based tool selection**, where the agent decides which tool to call dynamically
- Integrated **Model Context Protocol (MCP)** using FastMCP to connect the agent with an external data source
- Ensured the agent **retrieves real data instead of relying only on LLM knowledge**
- Generated **final responses using retrieved structured data + Gemini reasoning**
- Deployed the complete system on **Google Cloud Run** for real-world accessibility

Opportunities

- **How different is it from any of the other existing ideas?**

Uses MCP-based structured data retrieval instead of relying only on LLM-generated answers

- **How will it be able to solve the problem?**

Provides accurate, reliable, and instantly accessible medical information through a structured knowledge base

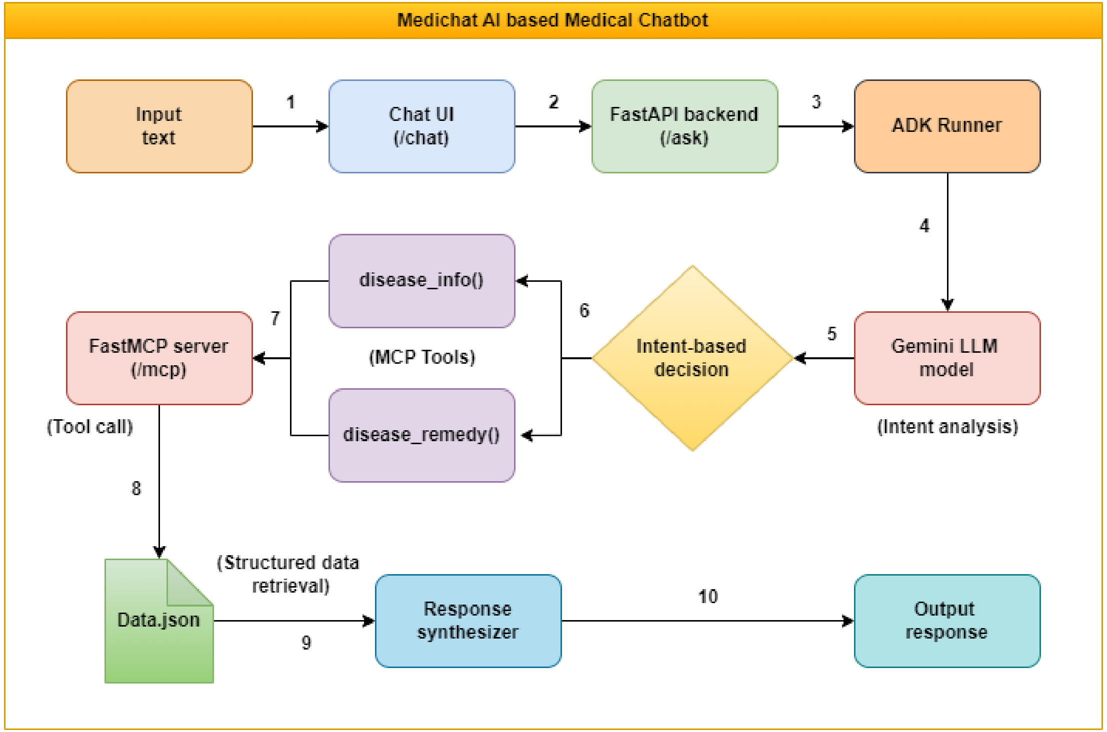
- **USP of the proposed solution**

Intent-driven tool calling with safety-first responses, ensuring relevant and trustworthy outputs

List of features offered by the solution

- AI-powered **medical assistant agent** for quick queries
- **MCP-based integration** with structured medical knowledge base
- **Two specialized tools**: disease information & remedies
- **Intent-based tool selection** for accurate responses
- **Structured and easy-to-understand outputs**
- **Safety-first responses with medical disclaimer**
- **Real-time interaction via chat UI**
- **Scalable deployment on Google Cloud Run**

Process flow diagram or Use-case diagram



Technologies used in the solution

- **Google ADK (Agent Development Kit)** — for building and orchestrating the AI agent
- **Gemini 1.5 Flash** — for fast and efficient LLM-based reasoning
- **FastMCP** — to expose tools as MCP-compatible endpoints
- **FastAPI + Uvicorn** — for backend API and request handling
- **Vanilla HTML, CSS, JavaScript** — for lightweight chat UI
- **Docker** — for containerization
- **Google Cloud Run** — for scalable, serverless deployment
- **Python 3.12** — core development language
- **Model Context Protocol (MCP)** — for standardized tool and data integration

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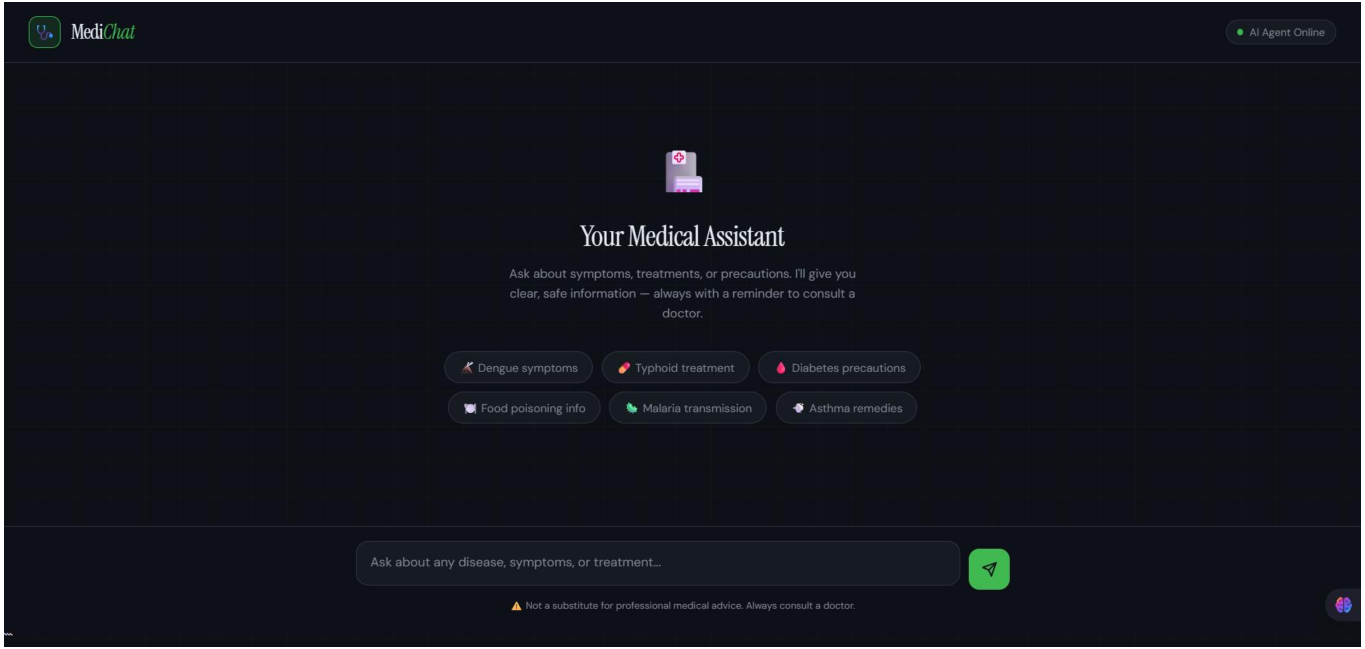
Snapshots of the prototype



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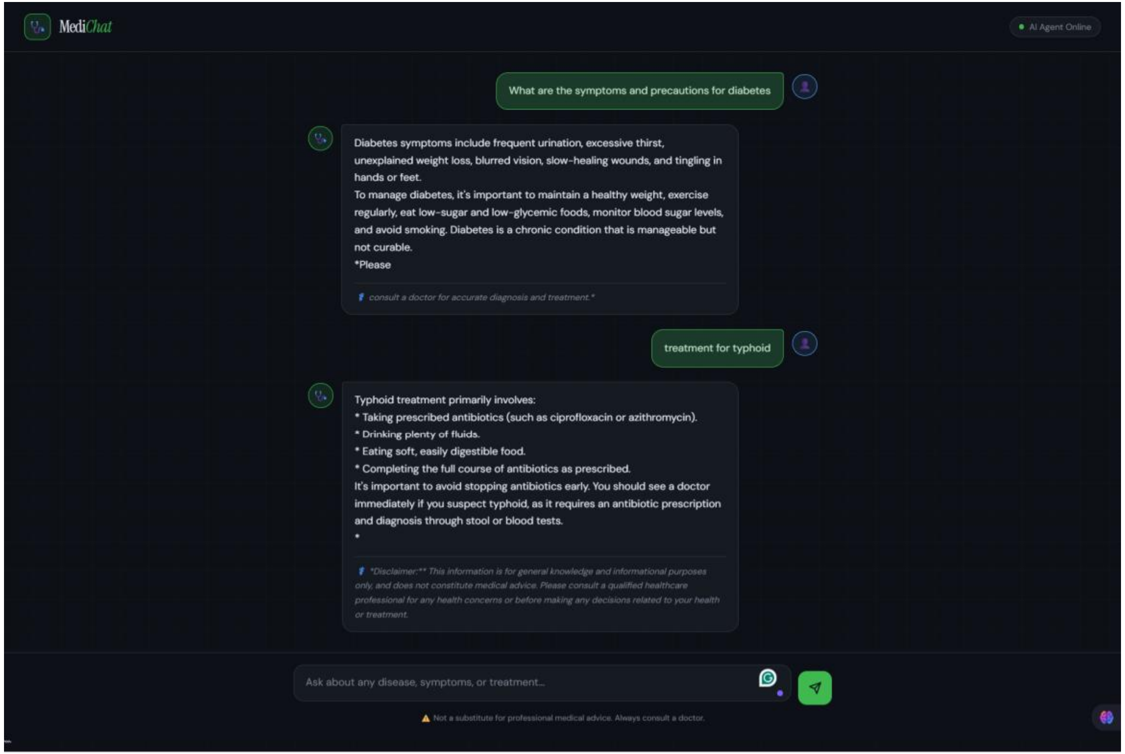

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Thank you